

Quiz 2 [25min]

1. **proposition** : a statement that is true or false.
2. **propositional variable** : a variable that represents a proposition.
3. **$\neg p$ (negation of p)** : the proposition with truth value opposite to the truth value of p .
4. **truth table** : a table displaying all possible truth values of propositions.
5. **$p \vee q$ (disjunction of p and q)** : the proposition " p or q ," which is true if and only if at least one of p and q is true.
6. **$p \wedge q$ (conjunction of p and q)** : the proposition " p and q ," which is true if and only if both p and q are true.
7. **$p \oplus q$ (exclusive or of p and q)** : the proposition " p XOR q ," which is true when exactly one of p and q is true.
8. **$p \rightarrow q$ (p implies q)** : the proposition "if p , then q ," which is false if and only if p is true and q is false.
9. **converse of $p \rightarrow q$** : the conditional statement $q \rightarrow p$.
10. **contrapositive of $p \rightarrow q$** : the conditional statement $\neg q \rightarrow \neg p$.
11. **inverse of $p \rightarrow q$** : the conditional statement $\neg p \rightarrow \neg q$.
12. **$p \leftrightarrow q$ (biconditional)** : the proposition " p if and only if q ," which is true if and only if p and q have the same truth value.
13. **tautology** : a compound proposition that is always true.
14. **contradiction** : a compound proposition that is always false.
15. **contingency** : a compound proposition that is sometimes true and sometimes false.
16. **consistent compound propositions / compound 생략 가능** : compound propositions for which there is an assignment of truth values to the variables that makes all these propositions true.
17. **satisfiable compound proposition / compound 생략 가능** : a compound proposition for which there is an assignment of truth values to its variables that makes it true.
18. **logically equivalent compound propositions / compound 생략 가능** : compound propositions that always have the same truth values.
19. **predicate** : part of a sentence that attributes a property to the subject.
20. **propositional function** : a statement containing one or more variables that becomes a proposition when each of its variables is assigned a value or is bound by a quantifier.
21. **domain (or universe) of discourse** : the values a variable in a propositional function may take.
22. **$\exists x P(x)$ (existential quantification of $P(x)$)** : the proposition that is true if and only if there exists an x in the domain such that $P(x)$ is true.
23. **$\forall x P(x)$ (universal quantification of $P(x)$)** : the proposition that is true if and only if $P(x)$ is true for every x in the domain.
24. **free variable** : a variable not bound in a propositional function.
25. **bound variable** : a variable that is quantified.
26. **scope of a quantifier** : portion of a statement where the quantifier binds its variable.
27. **argument** : a sequence of statements.
28. **argument form** : a sequence of compound propositions involving propositional variables.
29. **premise** : a statement, in an argument, or argument form, other than the final one.
30. **conclusion** : the final statement in an argument or argument form.
31. **valid argument form** : a sequence of compound propositions involving propositional variables where the truth of all the premises implies the truth of the conclusion.
32. **valid argument** : an argument with a valid argument form.
33. **rule of inference** : a valid argument form that can be used in the demonstration that arguments are valid.
34. **proof** : a demonstration that a theorem is true.
35. **vacuous proof** : a proof that $p \rightarrow q$ is true based on the fact that p is false.
36. **trivial proof** : a proof that $p \rightarrow q$ is true based on the fact that q is true.
37. **direct proof** : a proof that $p \rightarrow q$ is true that proceeds by showing that q must be true when p is true.
38. **proof by contraposition** : a proof that $p \rightarrow q$ is true that proceeds by showing that p must be false when q is false.
39. **proof by contradiction** : a proof that p is true based on the truth of the conditional statement $\neg p \rightarrow q$, where q is a contradiction.
40. **exhaustive proof** : a proof that establishes a result by checking a list of all possible cases.

41. **proof by cases** : a proof broken into separate cases, where these cases cover all possibilities.
42. **counterexample** : an element x such that $P(x)$ is false.
43. **nonconstructive existence proof** : a proof that an element with a specified property exists that does not explicitly find such an element.
44. **uniqueness proof** : a proof that there is exactly one element satisfying a specified property.
45. **onto function, surjection** : a function from A to B such that every element of B is the image of some element in A .
46. **one-to-one function, injection** : a function such that the images of elements in its domain are distinct.
47. Let $\mathbf{A} = [a_{ij}]$ be an $m \times k$ zero-one matrix and $\mathbf{B} = [b_{ij}]$ be a $k \times n$ zero-one matrix. Then the *Boolean product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} \odot \mathbf{B}$, is the $m \times n$ matrix with (i, j) th entry $c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \dots \vee (a_{ik} \wedge b_{kj})$
48. Let f be a function from the set A to the set B . The *graph* of the function f is $\{(a, b) \mid a \in A \text{ and } f(a) = b\}$.